# Xiaole Zhao

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I am currently an associate professor in the School of Computing and Artificial Intelligence, Southwest Jiaotong University (SWJTU), Chengdu, China. My research interests mainly include image restoration and reconstruction, deep learning and pattern recognition etc. Recently, I have been working on low-level computer vision tasks of medical images with deep learning techniques.

## EDUCATION

Sep. 2016 – Jun. 2020 **Biomedical Engineering**, *School of Life Science and Technology*, — **Ph.D.**
**University of Electronic Science and Technology of China (UESTC)**, Chengdu, China
**Advisor**: Prof. Mark Zou (Founder of Alltech Medical System Co., LTD) and Tao Zhang
Overall GPA: 3.56/4.0, Major GPA: 3.62/4.0

Sep. 2013 – Jun. 2016 **Software Engineering**, *School of Computer Science and Technology*, — **M. S.**
**Southwest University of Science and Technology (SWUST)**, Mianyang, China
**Advisor**: Prof. Yadong Wu
Overall GPA: 3.72/4.0, Major GPA: 3.70/4.0

Sep. 2009 – Jun. 2013 **Software Engineering**, *School of Computer Science and Technology*, — **B. S.**
**Southwest University of Science and Technology (SWUST)**, Mianyang, China
Overall GPA: 3.64/4.0, Major GPA: 3.66/4.0

## RESEARCH EXPERIENCE

Sep. 2025 – Now **Associate Professor**, Southwest Jiaotong University, Chengdu, China.
**Research**: Computer Vision, Image Processing, Deep Learning, Artificial Intelligence.

Jan. 2021 – Sep. 2025 **Assistant Professor**, Southwest Jiaotong University, Chengdu, China.
**Research**: Computer Vision, Image Processing, Deep Learning, Artificial Intelligence.

Jun. 2020 – Jan. 2021 **Associate Professor**, Southwest University of Science and Technology, Mianyang, China.
**Research**: Computer Vision, Image Processing, Deep Learning, Artificial Intelligence.

Sep. 2016 – Jun. 2020 **Ph.D Candidate**, University of Electronic Science and Technology of China, Chengdu, China.
**Research**: Medical Image Analysis, Magnetic Resonance Imaging, Deep Learning

Sep. 2016 – Jun. 2020 **Project Officer**, Alltech Medical System Co., LTD, Chengdu, China.
**Research**: Medical Image Analysis, Magnetic Resonance Imaging, Pattern Recognition, Deep Learning

## PROJECTS

[P1] **Youth Science Funds of National Natural Science Foundation of China**, **¥ 300,000**.
**Role**: Principal Investigator (PI) Jan. 2022 – Dec. 2024
**Project Title**: Research on Super-Resolution Enhanced Medical Image Diagnosis based on GANs ■

[P2] **General Program of Natural Science Foundation of Sichuan Province**, **¥ 150,000**.
**Role**: Principal Investigator (PI) Jan. 2026 – Dec. 2027
**Project Title**: Research on Spatial Resolution Enhancement Methods for Magnetic Resonance Imaging Assisted by Multimodal Ancillary Information ■

[P3] **Youth Science Funds of Natural Science Foundation Sichuan Province**, **¥ 100,000**.
**Role**: Principal Investigator (PI) Jan. 2022 – Dec. 2023
**Project Title**: Research on Joint Learning Methods of Medical Image Resolution Enhancement and Automatic Diagnosis and Recognition ■

[P4] **Fundamental Research Funds for the Central Universities**, **¥ 100,000**.
**Role**: Principal Investigator (PI) Jan. 2021 – Dec. 2022
**Project Title**: Research on Unsupervised Medical Image Super-Resolution Reconstruction for Clinical Diagnosis and Recognition ■

# PUBLICATIONS

**Overview** I have published more than 30 papers on reputable journals and conferences like TIP, TMM, TCSVT, TNNLS, INFFUS, TCE, CVMJ, CVPR, ICCV, ACM MM, ICME, ICASSP etc. Most of these papers focus on a low-level computer vision task, i.e., image super-resolution, image denoising, image segmentation etc., but also involve deep learning, deep clustering, pattern recognition, neuroimaging analysis and other related fields. My academic homepage is located at https://zxlation.github.io/xiaole.github.com.

**Journals**

[J1] **Xiaole Zhao**, Xinkun Wu, Xiangsong Jia, Xun Gong, Xiaobo Zhang, Tianrui Li. Scale-wise upsampling for efficient single image super-resolution. **IEEE Transactions on Consumer Electronics (TCE)**, 2026, 72(1): 670-681.

[J2] Yan Liu, Yan Yang, Yongquan Jiang, **Xiaole Zhao**, Zhuyang Xie. FABRF-Net: A frequency-aware boundary and region fusion network for breast ultrasound image segmentation. **Information Fusion (INFFUS)**, 2025, 123: 103299.

[J3] Pengzhan Chen, Shengdong Du, **Xiaole Zhao**, Jie Hu, Jingjing Li, Tianrui Li. DCTFormer: A dual-branch transformer with cloze tests for video anomaly detection. **IEEE Transactions on Multimedia (TMM)**, 2025, 27: 9022-9032.

[J4] Changlong Guo, Yan Yang, Yongquan Jiang, Xiaobo Zhang, **Xiaole Zhao**, Jie Wang. Joint multi-scale channel attention and multi-perception head for underwater object detection. **Big Data Mining and Analytics (BDMA)**, 2025, 8(6): 1335-1352.

[J5] Jin Gu, Xinbei Zha, Jiaming Zhang, **Xiaole Zhao**. Assisted diagnosis of neuropsychiatric disorders based on functional connectivity: A survey on application and performance evaluation of graph neural network. **Expert Systems with Applications (ESWA)**, 2025, 265: 125922.

[J6] Yunyang Wu, Xiaobo Zhang, **Xiaole Zhao**, Yimin Sun, Tianrui Li. segWCD: A new segmentation-based weak supervision neural network for building change detection. **Applied Intelligence (APIN)**, 2025, 55(2): no.147.

[J7] Xiaobo Zhang, Tao Wang, **Xiaole Zhao**, Dengmin Wen, and Donghai Zhai. Multitask-guided deep clustering with boundary adaptation. **IEEE Transactions on Neural Networks and Learning Systems (TNNLS)**, 2024, 35(5): 6089-6102.

[J8] **Xiaole Zhao**, Yulun Zhang, Yun Qin, Qian Wang, Tao Zhang, and Tianrui Li. Single MR image super-resolution via channel splitting and serial fusion network. **Knowledge-Based Systems (KBS)**, 2022, 246: 108669.

[J9] Haoqian Wang, Xiaowan Hu, **Xiaole Zhao**, Yulun Zhang. Wide weighted attention multi-scale network for accurate MR image super-resolution. **IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)**, 2022, 32(3): 962-975.

[J10] **Xiaole Zhao**, Xiafei Hu, Ying Liao, Tian He, Tao Zhang, Xueming Zou, Jinsha Tian. Accurate MR image super-resolution via lightweight lateral inhibition network. **Computer Vision and Image Understanding (CVIU)**, 2020, 201: 103075.

[J11] **Xiaole Zhao**, Huali Zhang, Yuliang Zhou, Wei Bian, Tao Zhang, Xueming Zou. Gibbs-ringing artifact suppression with knowledge transfer from natural images to MR images. **Multimedia Tools and Applications (MTAP)**, 2020, 79(45): 33711-33733.

[J12] Yun Qin, Nan Zhang, Yan Chen, Xiaojun Zuo, Sisi Jiang, **Xiaole Zhao**, Li Dong, Jianfu Li, Tao Zhang, Dezhong Yao, Cheng Luo. Rhythmic network modulation to thalamocortical couplings in epilepsy. **International Journal of Neural Systems (IJNS)**, 2020, 30(11): 2050014.

[J13] **Xiaole Zhao**, Yulun Zhang, Tao Zhang, and Xueming Zou. Channel splitting network for single MR image super-resolution. **IEEE Transactions on Image Processing (TIP)**, 2019, 28(11): 5649-5662.

[J14] **Xiaole Zhao**, Yadong Wu, Jinsha Tian, and Hongying Zhang. Single image super-resolution via blind blurring estimation and dictionary learning. **Neurocomputing**, 2016, 212: 3-11.

[J15] **Xiaole Zhao**, Yadong Wu, Jinsha Tian, and Hongying Zhang. Single image super-resolution via blind blurring estimation and anchored space mapping. **Computational Visual Media (CVMJ)**, 2016, 21(1): 71-85.

**Conferences**

[C1] **Xiaole Zhao**, Qingsong Pang, Xiaobo Zhang, Xun Xu, Xun Gong, Yan Yang, and Tianrui Li. Zero-shot image denoising via hybrid prior-guided pseudo sample generation. **IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)**, Accepted, 2026.

[C2] **Xiaole Zhao**, Tao Zhang, Yuwei Jin, Xiaobo Zhang, and Yan Yang. Segmentation-guided real hair removal based on simulated hair generation. **IEEE International Conference on Multimedia and Expo (ICME)**, Accepted, 2026.

[C3] Yan Zhou, **Xiaole Zhao**, Yuwei Jin. Image complexity assessment via detail-semantics synergy network. **IEEE International Conference on Multimedia and Expo (ICME)**, Accepted, 2026.

[C4] Chengxing Xie, Linze Li, **Xiaole Zhao**. Spatial dual feature aggregation network for lightweight image super-resolution. **International Joint Conference on Neural Networks (IJCNN)**, 2025: 1-8, DOI: 10.1109/IJCNN64981.2025.11227443.

[C5] Wenjia Yang, **Xiaole Zhao**, Xiaobo Zhang, Yan Yang, Tianrui Li. Hierarchical feature learning based on memory for skin lesion recognition. **International Joint Conference on Neural Networks (IJCNN)**, 2025: 1-8, DOI: 10.1109/IJCNN64981.2025.11227813.

[C6] Yaohua Li, **Xiaole Zhao**, Xiaobo Zhang, Yan Yang, Tianrui Li. Long-range multi-scale fusion for efficient single image super-resolution. **IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)**, 2025: 1-5, DOI: 10.1109/ICASSP49660.2025.10888924.

[C7] **Xiaole Zhao**, Linze Li, Chengxing Xie, Xiaoming Zhang, Ting Jiang, Wenjie Lin, Shuaicheng Liu, Tianrui Li. Efficient single image super-resolution with entropy attention and receptive field augmentation. **ACM International Conference on Multimedia (ACM MM)**, 2024: 1302-1310.

[C8] Xiangsong Jia, Xinkun Wu, Xiaobo Zhang, **Xiaole Zhao**. AwmFace: Adaptive weighting margin for deep face recognition. **International Joint Conference on Neural Networks (IJCNN)**, 2024: 1-8, DOI: 10.1109/IJCNN60899.2024.10651136.

[C9] **Xiaole Zhao**, Xinkun Wu. Lightweight image super-resolution with scale-wise network. **British Machine Vision Conference (BMVC)**, 2023: no.286.

[C10] Xiaoming Zhang, Tianrui Li, and **Xiaole Zhao**. Boosting single image super-resolution via partial channel shifting. **IEEE/CVF International Conference on Computer Vision (ICCV)**, 2023: 13223-13232.

[C11] Chengxing Xie, Xiaoming Zhang, Linze Li, Haiteng Meng, Tianlin Zhang, Tianrui Li, **Xiaole Zhao**. Large kernel distillation network for efficient single image super-resolution. **IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)**, 2023: 1283-1292.

[C12] Xiaowan Hu, Zhihong Liu, Ruijun Ma, Yuanhao Cai, Haoqian Wang, **Xiaole Zhao**, and Yulun Zhang. Pseudo 3D auto-correlation network for real Image denoising. **IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)**, 2021: 16175-16184.

[C13] Xiaowan Hu, Haoqian Wang, Yuanhao Cai, **Xiaole Zhao**, Yulun Zhang. Pyramid orthogonal attention network based on dual self-similarity for accurate MR image super-resolution. **IEEE International Conference on Multimedia and Expo (ICME)**, 2021: 1-6.

## PATENTS

[Patent 1] **Xiaole Zhao**, Linze Li, Jin Gu, and Shengdong Du. An entropy attention-based image super-resolution method, apparatus, and device. Patent No. ZL 2024 1 0772902.0, 2024.08.23, CNIPA.

[Patent 2] **Xiaole Zhao**, Yaohua Li, Jin Gu, Shengdong Du, and Zonghai Zhu. A multi-scale feature fusion-based image super-resolution method, apparatus, and device. Patent No. ZL 2024 1 0780901.0, 2024.08.30, CNIPA.

[Patent 3] **Xiaole Zhao**, Wenjia Yang, Shengdong Du, Zonghai Zhu, and Xiaobo Zhang. A hierarchical feature memory learning-based skin disease identification method. Patent No. ZL 2024 1 1423796.1, 2024.12.06, CNIPA.

[Patent 4] **Xiaole Zhao**, Xiaoming Zhang, Tianrui Li, and Xiaobo Zhang. A non-residual structure-based extremely deep image super-resolution reconstruction method. Patent No. ZL 2022 1 0793571.X, 2025.03.14, CNIPA.

[Patent 5] **Xiaole Zhao**, Tao Zhang, Shengdong Du, Jin Gu, Xiaobo Zhang, and Yan Yang. A hair removal network-based skin image classification method, apparatus, and product. Patent No. ZL 2024 1 1674166.1, 2025.11.04, CNIPA.

## ACADEMIC ACTIVITIES

### ★ Academic Service

I was invited to serve as the Collaboration Chair for PAKDD 2022 and as a reviewer for renowned academic journals such as TPAMI, TIP, TMI, TCSVT, TMM, TNNLS, CIM, TCSVT, as well as for top international conferences including CVPR, ICCV, ECCV, ACM MM, MICCAI, NeurIPS, and ICLR. I currently serve as a Youth Editorial Board Member for prestigious domestic journals such as Journal of Computer Applications and Journal of Naval Aviation University.

### ★ Membership

- ☐ Institute of Electrical and Electronics Engineers (**IEEE**) Since 2019
- ☐ China Computer Federation (**CCF**) Since 2021
- ☐ Association for Computing Machinery (**ACM**) Since 2023

### ★ Academic Conferences

- ☐ International Joint Conference on Neural Networks, Rome, Italy. 2025
- ☐ IEEE International Conference on Acoustics, Speech, and Signal Processing, Hyderabad, India. 2025
- ☐ ACM International Conference on Multimedia, Melbourne, Australia. 2024
- ☐ International Joint Conference on Neural Networks, Yokohama, Japan. 2024
- ☐ British Machine Vision Conference, Aberdeen, UK. 2023
- ☐ IEEE/CVF International Conference on Computer Vision, Paris, France. 2023
- ☐ IEEE/CVF Conference on Computer Vision and Pattern Recognition, Vancouver, Canada. 2023
- ☐ IEEE International Conference on Multimedia and Expo, Virtual Event. 2021
- ☐ IEEE/CVF Conference on Computer Vision and Pattern Recognition, Virtual Event. 2021
- ☐ International Society for Magnetic Resonance in Medicine, Sydney, Australia. 2020
- ☐ International Conference on Intelligence Science and Big Data Engineering, Nanjing, China. 2019
- ☐ Chinese Conference on Pattern Recognition and Computer Vision, Guangzhou, China. 2018
- ☐ International Conference on Computational Visual Media, Cardiff University, UK. 2016